Book of Abstracts



November 12, 2026 | 9 am - 6 pm

I2BC -CNRS , Gif-sur-Yvette 91400 , Auditorium 21

université PARIS-SACLAY

GRADUATE SCHOOL
Life Sciences
and Health

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Introduction

From Proteins to Genomes: New Computational Approaches Across Biological Scales

Biology produces more data than ever, from genomic sequences to three-dimensional protein structures. Making sense of it takes computational methods, and machine learning and AI now help us find patterns and make predictions that were out of reach not long ago.

This year's Junior Conference on Computational Biology is built around two themes. The morning session, Protein Folding and Evolution, looks at how protein structures can be predicted, compared and searched, and what they tell us about evolution. The keynote by Johannes Söding (Max Planck Institute for Multidisciplinary Sciences) will present recent tools for structure comparison and for modeling peptide-MHC complexes.

The afternoon session, GenOMIC Data Mining, focuses on getting biological knowledge out of very large genomic datasets, from millions of bacterial genomes to public RNA-seq data and the genetic causes of antimicrobial resistance. Antoine Limasset (CNRS, Université de Lille) will give the keynote on scaling sequence analysis to millions of bacterial genomes.

JC2B is organised by students, and we hope this day of talks, posters and internship fair gives everyone, from students to established researchers, a chance to share their work, talk freely and discover what others are working on.

Overview *on the* Organisation

The Junior Conference on Computational Biology 2026 was organised by students of the Master 2 program in Bioinformatics, Biostatistics and Applications of Artificial Intelligence (BIBS-IA) at Université Paris-Saclay.

The students worked collaboratively across teams on various aspects of the event, including website development, registration management, abstract selection, book of abstracts preparation, session chairing, communication and logistics.

Overview *on the* Organisation

I. Website and registration team

Developed the website, managed online registration and abstract submission forms, collected the information required for abstract selection and the book of abstracts, and continuously updated the event's programme.

2. Communication team

Promoted the conference through institutional emails and social media, followed up on registrations, and reached out to companies and research laboratories regarding internship opportunities for the internship fair.

3. Speaker management team

Defined criteria for abstract selection, reviewed and selected abstracts, notified accepted speakers, provided practical information to speakers and participants (such as the campus map), and collected speakers' presentations.

Overview *on the* Organisation

4. Logistics team

Collected keynote speakers' titles and abstracts, coordinated catering services for all attendees, managed keynote travel arrangements, and oversaw the event's budget.

5. Event organisation team

Set up the conference venue, managed audio-visual equipment and internship fair tables, and chaired the conference sessions.

6. Organisation team

Managed attendee reception, designed and distributed badges, prepared the book of abstracts, and created a quiz based on the conference presentations.

Conference program

9h00 — **Opening Speech**

MORNING SESSION : Protein Folding And Evolution

9h20 - 9h45 — **Paul Roginski - CNRS -**
A robust proteome of LECA using structural phylogenomics
History of eukaryotes • Protein • Database creation

9h45 -10h10 — **Lucas Vitoriano De Queiroz Lira - I2BC -**
How much does AlphaFold2 rely on coevolution?
AlphaFold • Coevolution

10h10 - 10h35 — **Charlotte Chapellier - Ecole des Ponts -**
Can data-driven collective variables yield an efficient reduced representation to reconstruct the Heat-Shock Protein 90α N-Terminal Domain free energy landscape?
Molecular dynamics

10h35 - 11h20 — **Coffee Break - POSTER SESSION & INTERNSHIP FAIR**

11h20 - 11h45 — **Roland Faure - Institut Pasteur -**
Building a search engine for proteins
Structure protein prediction • Protein comparison

11h45 - 12h30 — **KEYNOTE : Pr. Dr. Johannes Söding**
- Max-Planck-Institute for Multidisciplinary Sciences-
Fast structure comparison with Foldseek2 and LoL-align and a toolbox for peptide-MHC complex modeling and binding prediction
Computational structural biology • Foldseek2 • LoL-align

12h30 - 14h00 — **Lunch Break**

Conference program

AFTERNOON SESSION : genOMIC Data Mining

14h00 - 14h45

KEYNOTE : Dr. Antoine Limasset

- Lille University , CRIStAL Laboratory

Scaling Sequence Analysis to Millions of Bacterial Genomes

Genomic Databases • Algorithms • Sequence Analysis • Scalability • Indexing.

14h45 - 15h10

Kristen Curry - Institut Pasteur -

Unconstrained identification of candidate genome-modifying loci

Genome • Tool development

15h10 - 15h35

Fiona Hak - I2BC -

Metappuccino: scalable reconstruction of metadata for systematic exploration of public RNA-seq dat

Metadata • RNA-Seq • Open DB • NLP

15h35 - 16h05 **Coffee Break - POSTER SESSION & INTERNSHIP FAIR**

16h05 - 16h30

Mario Favre-Moiron - Institut Curie - Servier -

Trans-pQTL Mapping via a Data-Driven Protein Clustering Approach

Loci/Protein Link • Genomics & Proteomics • Statistical Tools • Translational Research • Multi-omics Study

16h30 - 16h55

Naama Wagner - LISN -

Predicting and identifying the genetic cause of antimicrobial resistance from the full genome sequences using Large Language Models

Antimicrobial resistance • Language Models • Interpretable AI

16h55 - 17h20

Thomas Marino - INRAE -

Two Linkers as drivers of Mfd reshaping

Drug target • DNA damage • Sequences

17h20 **Closing Speech**

Speakers *and* Abstracts

Speaker
Paul Roginski

Laboratory
CNRS

Subject

A robust proteome of LECA using structural phylogenomics

History of eukaryotes • Protein • Database creation

Abstract The emergence of eukaryotes, with their complex cellular organization, was one of the major events in the history of life. Their diversity in morphologies, metabolisms, and lifestyles is unparalleled. Yet, all eukaryotes share a common cellular chassis inherited from their last common ancestor (LECA). What was LECA's genetic legacy? Was it a simple cell with just a few housekeeping genes and mitochondria, or a more complex one that has given 'us' a broad set of proteins? To address this question, we built an unprecedented collection of proteins, covering the largest diversity of eukaryotes to date—including neglected protists. For the first time, we applied structure-based approaches to detect orthogroups of proteins tracing back to LECA, producing the most robust core proteome of LECA yet. This might help us understand how this eukaryotic adventure began...

Speaker

Lucas Vitoriano De Queiroz Lira

Laboratory

I2BC

Subject

How much does AlphaFold2 rely on coevolution?

AlphaFold • Coevolution

Abstract

AlphaFold2 (AF2) accuracy depends on the multiple sequence alignment (MSA), but how much of that dependence comes from coevolutionary signal remains unresolved. We progressively shuffled residues within MSA columns, which destroys covariation while preserving conservation, on Pfam families with no structures in AF2's training set. Mean TM-score to the experimental structure falls from 0.68 with the intact MSA to 0.31 at full shuffling, close to single-sequence inference. Coevolution therefore accounts for nearly all of the benefit AF2 draws from the MSA. The damage is not uniform: long-range contacts are the most affected, and predicted structures become less compact. These results help clarify how AF2 uses coevolutionary signal and where in the structure it matters most.

Speaker

Charlotte Chapellier

Laboratory

Ecole des Ponts

Subject

Can data-driven collective variables yield an efficient reduced representation to reconstruct the Heat-Shock Protein 90α N-Terminal Domain free energy landscape?

Molecular dynamics

Abstract In molecular dynamics, transitions between conformations are rare events, posing significant challenges for sampling. Enhanced sampling methods, such as the extended Adapted Biasing Force (eABF), utilize collective variables (CVs) to capture the slow components of these transitions. While intuitive selection of CVs can sometimes be effective, it often fails to capture critical transitions. Building on the work of my predecessor, this study explores a novel approach using autoencoders combined with Linear Discriminant Analysis (LDA) to identify optimal CVs. By validating our methods on semi-supervised datasets of a protein or an RNA, we aim to maximize the likelihood of obtaining effective CVs for various therapeutic targets, thereby improving the efficiency and accuracy of enhanced sampling methods in molecular dynamics.

Speaker

Roland Faure

Laboratory

Institut Pasteur

Subject

Building a search engine for proteins

Structure protein prediction • Protein comparison

Abstract

In this talk, I will describe how we built Logan-Protein-Search, a search engine that finds proteins similar to a query in Logan-proteins, the largest existing protein database. Until recently, proteins were compared in two standard ways: aligning their amino-acid sequences, or folding them and aligning their three-dimensional structures. Structural alignment is the gold standard for quality, but becomes extremely expensive on large protein collections, while sequence comparison sometimes fails to detect highly similar proteins. Logan-Protein-Search uses Protein Language Models —the amino-acid equivalents of LLMs—to encode each protein as a vector, and retrieves the nearest neighbors of a query, much as web search engines do.

KEYNOTE

Speaker

Pr. Dr. Johannes Söding

Laboratory

Max-Planck-Institute for Multidisciplinary Sciences

Subject

Fast structure comparison with Foldseek2 and LoL-align and a toolbox for peptide-MHC complex modeling and binding prediction

Computational structural biology • Foldseek2 • LoL-align

Abstract I will first give an update on the fast protein structure search tool Foldseek2, including a new 12-state alphabet complementary to 3D. Whereas the focus in developing Foldseek was on speed, the goal in developing LoL-align was to attain maximum accuracy alignments and homology detection sensitivity. Finally, changing topics, I will describe our efforts to support computational and experimental immunologists with a toolkit for modeling peptide-MHC complexes, predicting binding to arbitrary MHC alleles, and designing peptide antigens, for instance for improved cancer vaccines.

KEYNOTE

Speaker

Dr. Antoine Limasset

Laboratory

CRIStAL

Subject

Scaling Sequence Analysis to Millions of Bacterial Genomes

Genomic Databases • Algorithms • Sequence Analysis • Scalability • Indexing

Abstract Genomics has entered an era of unprecedented scale, leaving us awash in sequence data but often unable to take full advantage of what is already available. Public sequencing databases are a genomic goldmine, from the vast archive of raw reads in the Sequence Read Archive to Logan, which assembles millions of sequencing datasets, or AllTheBacteria, which organises millions of standardized bacterial genomes. Algorithms and data structures are critical to unlocking this wealth of data, making it possible to search, compare, and extract relevant information from massive sequence collections. This keynote therefore explores sequence analysis from the most computationally expensive approaches to the most scalable. We will progress from alignment and mapping to comparisons based on shared k-mers, then examine how minimizers reduce the sequence content retained and how sketching and indexing enable searches across massive collections. The presentation develops a high-level understanding of the ideas behind these methods and the trade-offs they offer. We will examine what each approach can accomplish, its cost in time and memory, and the information it preserves or discards. Understanding these tradeoffs allows us to match the method to the biological question, avoiding both expensive analyses that add no insight and fast ones that cannot provide an answer.

Speaker

Kristen Curry

Laboratory

Institut Pasteur

Subject

Unconstrained identification of candidate genome-modifying loci

Genome • Tool development

Abstract Before CRISPR became a dominant gene-editing technology, its loci were first recognized as an unusual repeat pattern in bacterial genomes. Our software, Kiwi, aims to replicate this type of discovery by leveraging pangenome graph traversal and multiple sequence alignment entropy for a homology-free search. Performance has been validated on known genome-modifying systems, CRISPR and diversity-generating retroelements (DGR), where it successfully detects signatures, improves recall over state-of-the-art methods, and additionally flags instances that narrowly miss classification criteria as candidates for further investigation. Furthermore, Kiwi reveals loci with unique diversification patterns in unexpected genome contexts, providing compelling motivation for a targeted exploration of these overlooked regions of the bacterial genome.

Speaker

Fiona Hak

Laboratory

I2BC

Subject

Metappuccino: scalable reconstruction of metadata for systematic exploration of public RNA-seq data

Metadata • RNA-Seq • Open DB • NLP

Abstract Public RNA-seq data are difficult to reuse at scale because SRA metadata are sparse and inconsistent. Metappuccino reconstructs 15 metadata categories from structured records and free text by combining deterministic NLP, a LoRA-fine-tuned Mistral-7B model, ontology-based normalization and consistency checks. It outperformed larger general-purpose LLMs in benchmarks. Applied to one million human RNA-seq runs (2012–2025), it increased annotation coverage by >82 percentage points for organ, >77 for biopsy site, ~58 for disease and >43 for cell type. Each run also receives a confidence score based on cross-field biological consistency. The resulting open dataset supports reliable cohort selection and large-scale transcriptomic exploration, while the approach is being extended to other species.

Speaker

Faure-Moiron Mario

Laboratory

Institut Curie - Servier

Subject

Trans-pQTL Mapping via a Data-Driven Protein Clustering Approach

Loci/Protein Link • Genomics & Proteomics • Statistical Tools • Translational Research • Multi-omics Study

Abstract Protein Quantitative Trait Loci (pQTLs) link genetic variation and disease biology. However, standard studies lack statistical power to detect distal (trans)-pQTLs, and current recovery methods rely on prior knowledge. Here, we present a strategy to uncover trans-pQTLs via "co-pleiotropy": the overlap of proteins associated with variant pairs. Applying Rank-Rank Hypergeometric Overlap (RRHO) to the UKB-PPP and deCODE datasets, we compared variant association profiles to identify functional protein clusters without a priori assumptions. By aggregating subtle signals, we detect biologically relevant sub-threshold variants that recapitulate known protein-protein interaction networks. This framework offers a powerful tool to decode the post-transcriptional landscape of GWAS loci.

Speaker

Naama Wagner

Laboratory

LISN

Subject

Predicting and identifying the genetic cause of antimicrobial resistance from the full genome sequences using Large Language Models

Antimicrobial resistance • Language Models • Interpretable AI

Abstract Emergence of antibiotic resistant bacteria is a pressing issue in modern day medicine. To fight it and develop new drugs for new resistance mechanisms we have to detect the genetic cause of the resistance. Interpretable artificial intelligence models that predict the resistance phenotype from the full genome sequences are ideal for the task. To train these models, I use a genomic LLM, Evo, to produce embeddings from the genomic sequences. I embedded thousands of P. aeruginosa genomes from the Bakta database that were tested for different antibiotics, and trained classifiers to predict the resistance phenotype using these embeddings. By training a Positive-Unlabeled classifier on genomic windows, where windows from susceptible strains were labeled positive and windows from resistant strains were unlabeled, I was able to identify the windows harboring the relevant genes in the resistant strains.

Speaker
Thomas Marino

Laboratory
INRAE

Subject

Two Linkers as drivers of Mfd reshaping

Drug target • DNA damage • Sequences

Abstract Amid the urgent global crisis of antimicrobial resistance, the bacterial protein Mfd has emerged as an innovative drug target. Mfd is a key virulence factor that repairs DNA damage, caused by nitric oxide produced by macrophages during infection. As a non-essential transcription factor unique to bacteria, Mfd recognizes RNA polymerase stalled at non-coding lesions driving to recruit the nucleotide excision repair machinery, thereby promoting bacterial survival. We aim to interfere with the extensive three-dimensional remodeling Mfd undergoes to engage in functional partnerships. Combining sequence evolution analyses, molecular dynamics simulations, and targeted substitutions, we characterize dynamic patterns driving remodeling along key conformational states. Notably, we identified two long linkers flanking the central RNA polymerase–binding domain that may couple their motions to orchestrate the global Mfd.

Posters *and* Abstracts

Speaker
Cherif SEDDIK

Laboratory
INSERM U981 -
Gustave Roussy

Subject

Contribution of the dark transcriptome to immunotherapy response in melanoma

Abstract Melanoma is a malignant tumor that can metastasize and acquire resistance to anticancer treatments. Immune checkpoint blockade has significantly improved the survival of patients with metastatic melanoma. Yet half of the patients treated with this immunotherapy do not benefit from durable responses. Markers derived from RNA-sequencing are promising for potential use in clinical practice to predict response to immunotherapy. Standard approaches rely on the quantification of protein-coding transcripts, which represent a small fraction of the true diversity of RNA transcripts. This work aims explore a cohort of 163 patients to identify markers derived from the whole transcriptome, in particular, markers derived from RNAs that are little studied, referred to as the dark transcriptome. Using a k-mer based approach, we show novel markers associated with response to immunotherapy, inaccessible using standard approaches.

Speaker

Julie Fares

Laboratory

ITODYS

Subject

AGAPE 2.0 - Leveraging Machine Learning and Generative Approaches to Design Selective G4 Ligands

Abstract G-quadruplexes (G4s) have emerged as promising therapeutic targets, yet identifying active ligands remains challenging due to the vast chemical space and costly experimental screening. We present Deep-AGAPE, an open-source machine learning framework designed to predict G4 stabilization using molecular descriptors and experimental FRET melting data. Optimized XGBoost and deep neural network models achieved F1-scores of 0.848 and 0.810, respectively. Complementing this predictive approach, G4PT integrates generative artificial intelligence to design novel potential G4 ligands, achieving 61.5% molecular validity and approximately 94% novelty. Together, these frameworks combine ligand activity prediction and molecular generation in an accessible, reproducible workflow supported by interactive web platforms to accelerate G4-targeted drug discovery.

Speaker

Ines Malleval

Laboratory

Cancer Data Science Unit I36I-INSERM

Subject

Large Real World Cohort Validates Baseline and Early On-Treatment Biomarkers for Outcome Prediction in ICI Treated Metastatic Melanoma

Abstract Metastatic melanoma survival has been radically extended by immune checkpoint blockers but resistance occurs for half patients. Here, we hypothesized early on-treatment routine markers can enhance survival and response prediction together with baseline and clinical markers. We propose one model with baseline markers and also with early on-treatment routine markers on a 704 patients discovery cohort (Cox models are fitted in survival and logistic regression in response). We compare our models to 11 state of the art (STOTA) models on discovery and external validation cohorts. STOTA include baseline markers only, by adding early on-treatment markers correspondent to baseline markers, we compare gain in performances of on-treatment and baseline vs baseline markers for each models pairs. Our models beat baseline STOTA models. For each pairs, on-treatment markers improve performances and likelihood ratio test are significant.

Speaker

Andrea TARABINI

Laboratory

CQSB - UMR7238

Subject

Impact of Structural Variants on Genome and Proteome Evolution in Saccharomyces cerevisiae

Abstract Structural Variants (SV) are genomic rearrangements affecting DNA on a large scale. By altering gene dosage, disrupting coding sequences, or creating gene fusions, SVs affect gene function. Yet their functional and evolutionary impacts remain poorly understood. Leveraging the S. cerevisiae Reference Assembly Panel (ScRAP) pangenome and our upgraded SV-caller MUM&Co, we comprehensively identified SVs in the species population. By integrating them into the pangenome annotation we will assess their impact on gene repertoire evolution. Combining strain-specific annotations and proteomics data will reveal the effects of SVs on protein abundance and uncover SV-derived proteins. Structural properties of newly emerged protein domains will highlight SVs contribution to protein folds evolution. Ultimately, associations between SVs and phenotypic traits will be tested to provide insight into their adaptive significance.

Speaker

Samuel Gardelle

Laboratory

LIX

Subject

FPT learning and sampling of DCA models

Abstract RNA structure modeling greatly benefits from the availability of structural homologs, associated with the presence of coevolving positions in multiple alignments. Direct Coupling Analysis (DCA) is a statistical framework leveraging significant covariations between positions. Edge-activated DCA (eaDCA) was recently proposed as a way to target sparser and more data-efficient models using a simple greedy algorithm for learning, typically implemented with Markov Chain Monte Carlo techniques. In this work, we revisit the learning procedure in a rigorous algorithmic setting, circumventing the #P-hardness of the computation by devising exact fixed-parameter algorithms for the treewidth parameter. We empirically observe that eaDCA models are associated with moderate treewidth values, and validate the practicality of our implementation on 41 structured non-coding RFAM families.

Speaker

Meuret Louis

Laboratory

UPC-BFA

Subject

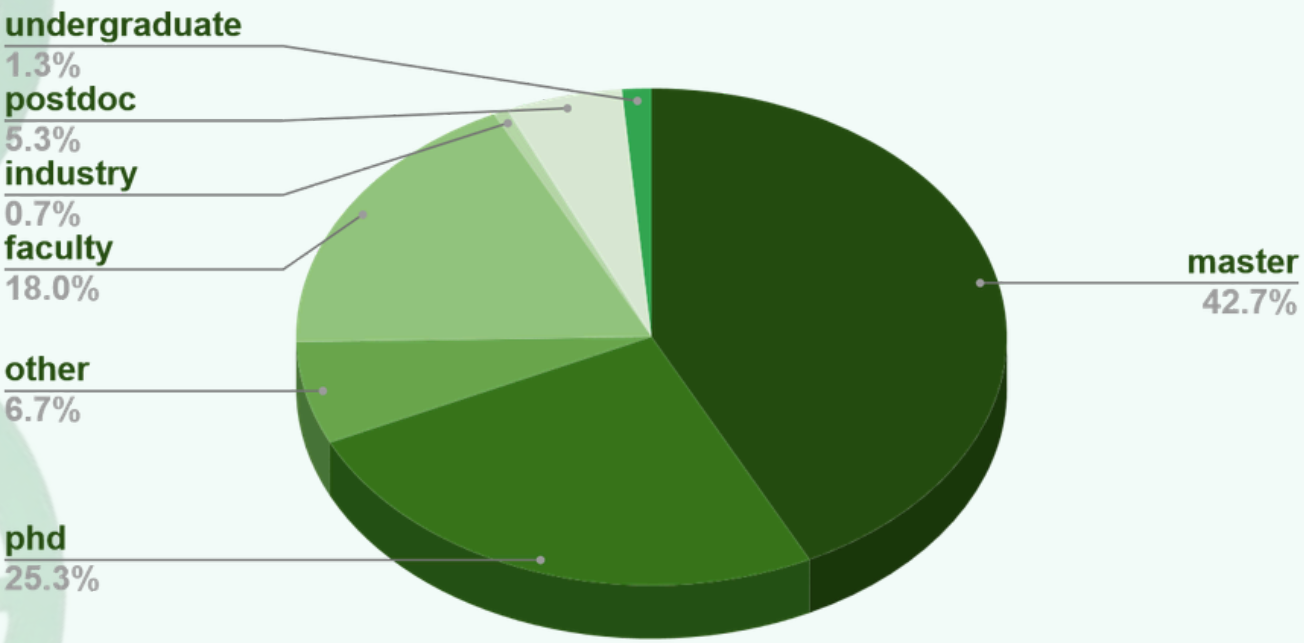
Molecular dynamics simulations of the Frameshifting Stimulatory Element of the SARS-CoV2 for drug design

Abstract Study of the Frameshifting Stimulatory Element using biaised molecular dynamics simulation to explore different conformational states, learn the folding pathways and try to identify small molecules that could interfere on those folding pathways.

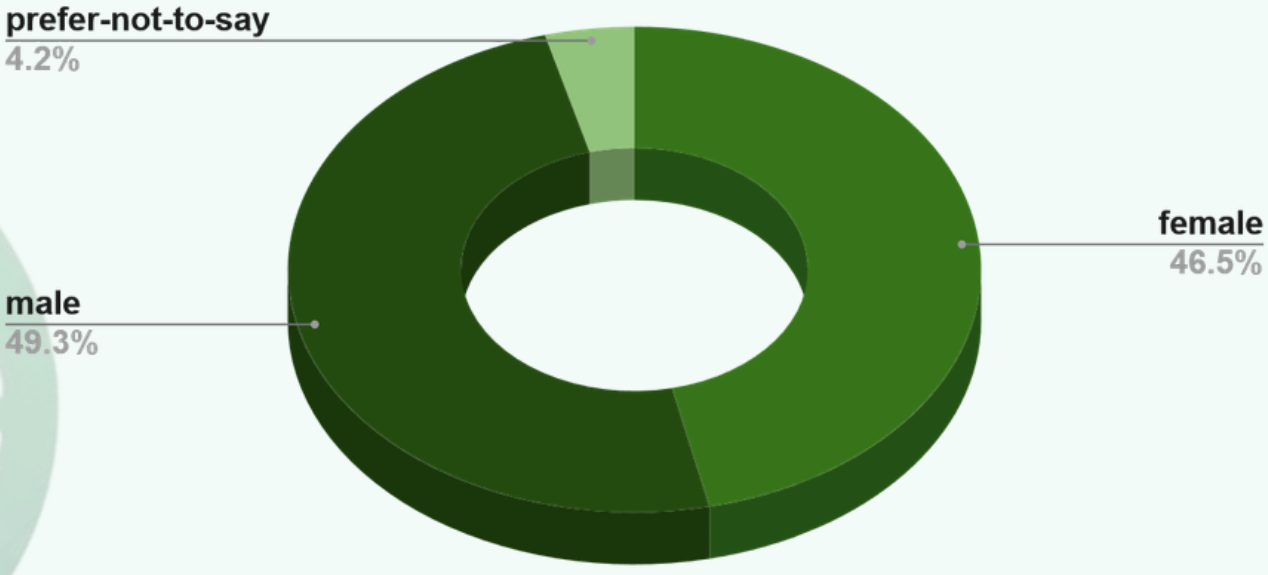
Some Statistics

150 participants
19 Universities & Schools
16 laboratories & companies

Participants' Academic Level

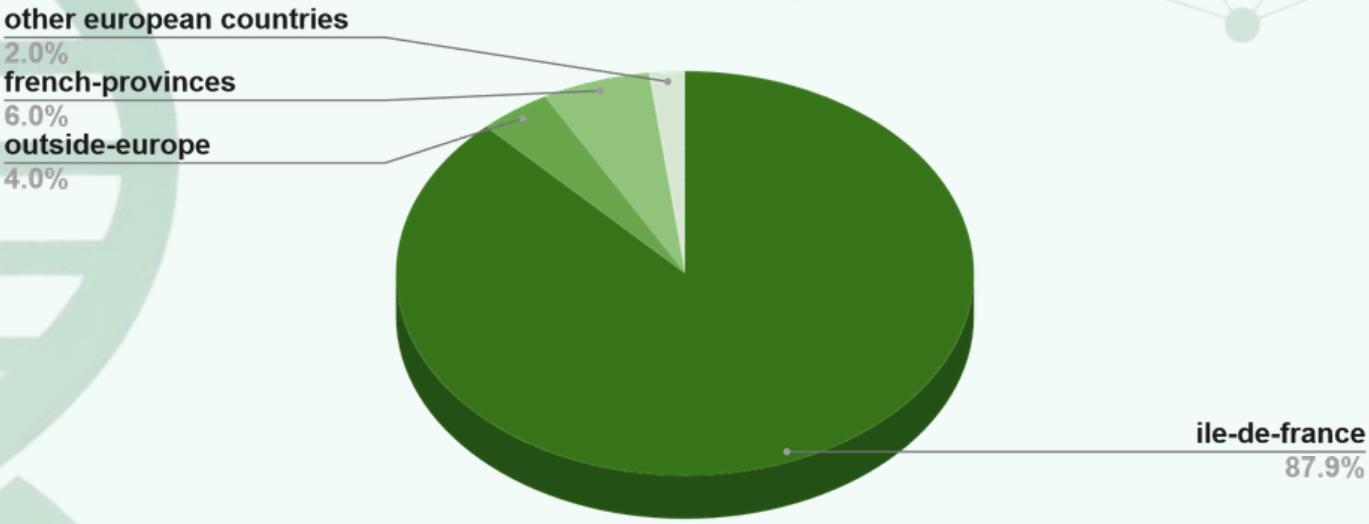


Participant Gender Distribution

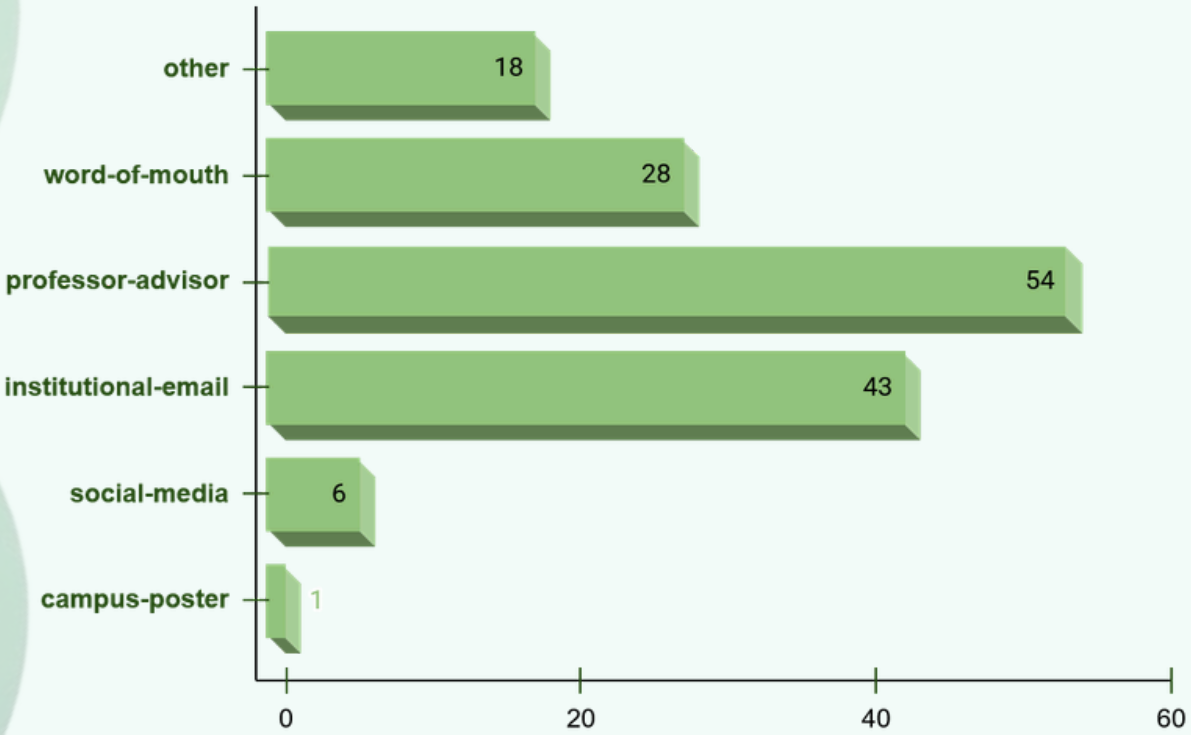


Some Statistics

Participants' Location Breakdown



How Participants Discovered the Event



Acknowledgements

As the JC2B 2026 organising committee, we would like to sincerely thank our faculty coordinators Anne Lopes and Daniel Gautheret for their consistent guidance and assistance in making this conference possible.

We would also like to extend our appreciation to our partners, namely, Université Paris-Saclay, The Institute of Integrative Cell Biology (I2BC), Living Machines At Work (LM@W), and Graduate School Life Sciences and Health (GS LSH), for their sponsorship.

Finally, we are grateful to the Master 2 BIBS-IA students for their investment, enthusiasm and teamwork, as well as to the speakers and participants, in ensuring a successful Junior Conference on Computational Biology 2026.



Edited by the JC2B Editorial Committee

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